

# **Virtual University of Pakistan**

**Che201**

**Physical chemistry**

**Midterm past papers solved**

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### Define Physical Chemistry?2

Ans: Physical chemistry is the study of the underlying physical principles that govern the properties and behavior of chemical systems.

### Define inions solutions?marks 2

Ans:

### Applications of equilibrium constant?3 marks

Ans: 1. The magnitude of the equilibrium constant,  $K$ , indicates the extent to which a reaction will proceed:

If  $K$  is a large number, it means that the equilibrium concentration of the products is large. In this case, the reaction as written will proceed to the right (resulting in an increase in the concentration of products)

If  $K$  is a small number, it means that the equilibrium concentration of the reactants is large. In this case, the reaction as written will proceed to the left (resulting in an increase in the concentration of reactants).

Knowing the value of the equilibrium constant,  $K$ , will allow us to determine:

The direction a reaction will proceed to achieve equilibrium

The ratios of the concentrations of reactants and products when equilibrium is reached

#### 2. Predicting the Direction of a Reaction

If  $Q = K_c$ , then the system is already at equilibrium

If  $Q > K_c$ , then essentially we have too much product and the reaction will proceed to the left (to reduce the concentration of product and increase the concentration of reactant)

If  $Q < K_c$ , then essentially we have too little product and the reaction will proceed to the right (to produce more product and decrease the concentration of reactant)

#### . Calculation of the Equilibrium Concentration of a Reactant or Product

Many types of equilibrium problems deal with determining how much of a product (or reactant) we will have once a reaction reaches equilibrium.

#### 4. Solving equilibrium concentrations of all components in a reaction

Sometimes an equilibrium problem will provide the value for the equilibrium constant and the initial concentration of all species

**Why ideal gas equation failure? Also describe corrections? 5 marks**

Ans: Thus we get the ideal gas equation:

$$pV = nRT$$

for ideal gas it is assumed as

Ideal gas particles occupy negligible volume

Ideal gas has negligible intermolecular interactions

Ideal gas equation shows deviations from

Avogadro's Hypothesis

Boyle's Law

Charles' Law

So it's called

Failures of ideal gas equation

**Write properties of colloids? 5**

Ans: Properties of the Colloids

- The particles of dispersant are between about 500 nm to 1 nm in diameter.
  - The mixture does not separate on standing in a standard gravity condition. (One 'g'.)
- The mixture does not separate by common filter, but might be filterable by materials with a smaller mesh.
- The mixture is not necessarily completely homogeneous, but usually close to being so.
- The mixture may appear cloudy or almost totally transparent, but if you shine a light beam through it, the pathway of the light is visible from any angle. This scattering of light is called the Tyndall effect
- There usually is not a definite, sharp saturation point at which no more dispersant can be taken by the dispersing agent.

- The dispersant can be coagulated, or separated by clumping the dispersant particles with heat or an increase in the concentration of ionic particles in solution into the mixture.
- There is usually only small effect of any of the colligative properties due to the dispersant.